MIL-R-36987
23 July 1973

MILITARY SPECIFICATION

RETRACTOR, TRACHEAL, HUPP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tracheal retractors of the Hupp design suitable for retraction of trachea in surgical procedures.

1.2 Classification. The general operating retractors shall be of the following types as specified (see 6.2):

Type I	-	1 sharp prong
Type II	-	3 sharp prongs

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Set-Up.

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Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The general operating retractor shall be in accordance with the applicable requirements of GG-I-526, and the following option exercises under that specification.

3.2 Material. The general operating retractor shall be fabricated from class 3 austenitic stainless steel, type 302, or 304 or class 4 martensitic stainless steel, type 420.

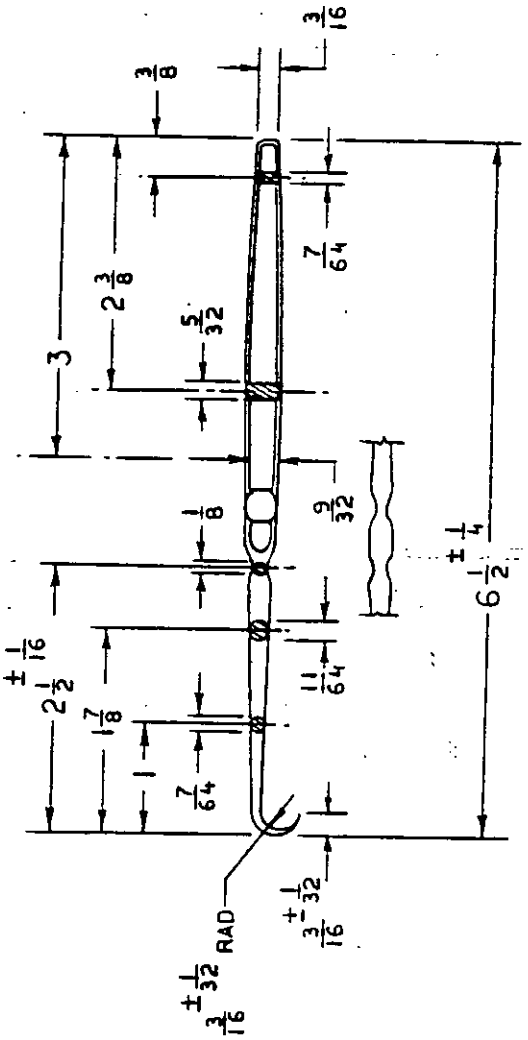
3.3 Hardness. After proper stress relieving, the Rockwell hardness shall be B75 to B85 for class 3 and shall be C40 to C45 for class 4.

3.4 Style, design, and dimensions. Shall be of one-piece construction and shall be in accordance with Figure 1 for Type I, or Figure 2 for Type II.

3.5 Finish. The finish shall be mirror, or mirror and satin.

3.6 Identification marking. In addition to GG-I-526 requirements, tracheal retractors of foreign manufacture shall also be marked "stainless".

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TYPE 1

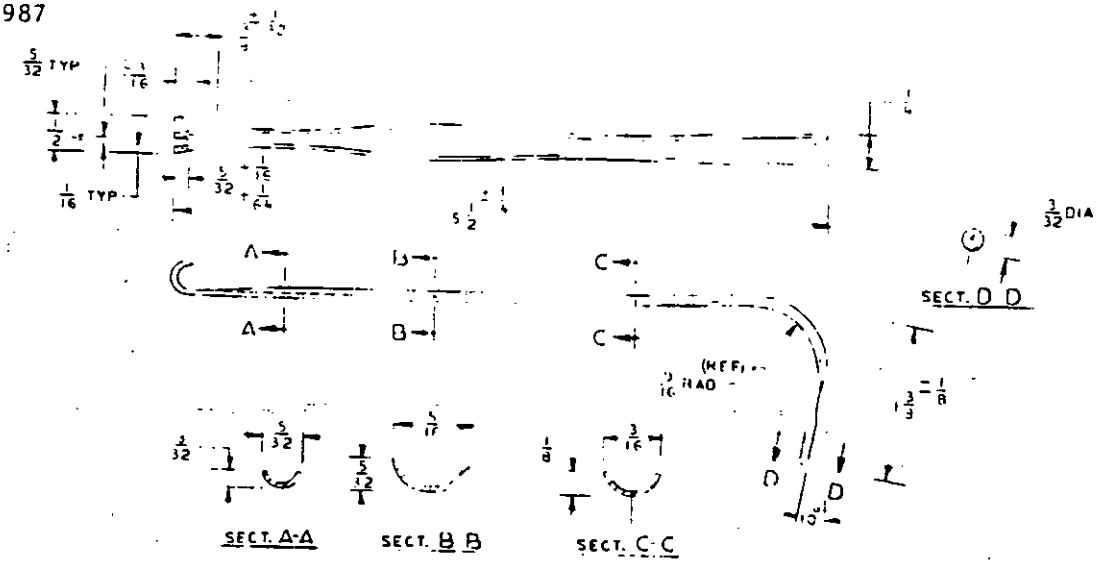
NOTES
UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES AN TOLERANCE IS ± 1/32 IN.

TOLERANCES INDICATED DELINEATE ACCEPTABLE EXTREME DIMENSIONS AND ARE NOT INTENDED TO PERMIT VARIATIONS WITHIN A SINGLE INSTRUMENT WHICH WILL SACRIFICE UNIFORMITY, SYMMETRY, AND FIRST QUALITY WORKMANSHIP.

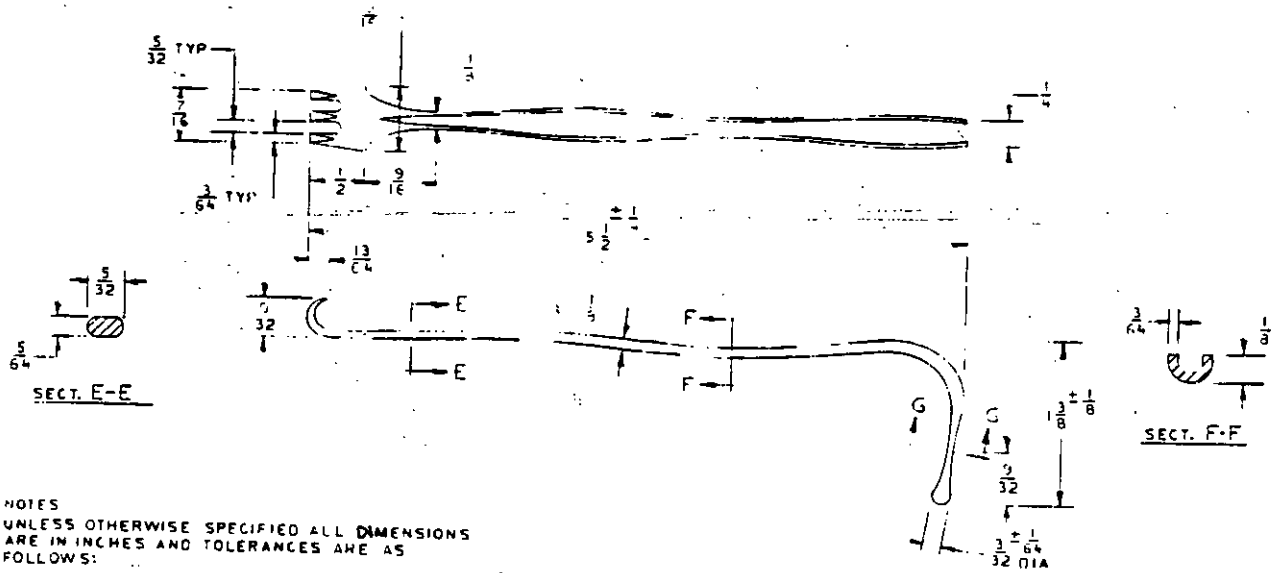
RETRACTOR, TRACHEAL, HUPP, 1 SHARP PRONG

FIGURE 1

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TYPE II-STYLE 1



NOTES
UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES AND TOLERANCES ARE AS FOLLOWS:

DIMS. UP TO 1/8 INCH ± 1/64 IN.
DIMS. OVER 1/8 INCH TO 3/8 INCH ± 1/32 IN.
DIMS. OVER 3/8 INCH ± 1/16 IN.
ANGLES ± 2°

TOLERANCES INDICATED DELINEATE ACCEPTABLE EXTREME DIMENSIONS AND ARE NOT INTENDED TO PERMIT VARIATIONS WITHIN A SINGLE INSTRUMENT WHICH WILL SACRIFICE UNIFORMITY, SYMMETRY, AND FIRST QUALITY WORKMANSHIP.

TYPE II-STYLE 2

RETRACTOR, TRACHEAL, HUPP, 3 SHARP PRONGS

FIGURE 2

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Shall be in accordance with GG-I-526.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the stainless steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination purposes shall be one retractor.

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
For visual examination		
Major defect	II	1.0
Minor defect	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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Table II. Sampling for Tests

Component (unit of product)	Characteristic	Requirement	Test Procedure	Inspection level	AQL (percent defective)
1-Retractor	Composition analysis	3.2 and GG-I-526	GG-I-526	*	1.0
1-Retractor	Hardness	3.3 and GG-I-526	GG-I-526	S-2	1.0
1-Retractor	Dimensions	3.4	--	S-2	2.5
1-Retractor	Copper sulfate	GG-I-526	GG-I-526	S-2	1.0
1-Retractor	Boil test	GG-I-526	GG-I-526	S-2	1.0

* Sufficient amount from each lot of each metal used in the construction of the retractor shall be selected for composition analysis.

4.3 Examination. The retractor shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Table I and III. Examination, however, shall not be restricted to defects listed in Table III. Any retractor in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for test sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Style and design not as specified.
102	Instrument not constructed as specified.
103	Instrument not free from sharp edges, rough areas, burrs, cracks, grind marks, or overlaps.
Minor	
201	Finish not as specified.
202	Instrument not permanently and legibly marked with either the name or trademark of the contractor.

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4.4 Tests. Tests shall be conducted to determine compliance with Specification requirements in accordance with GG-I-526 and as specified herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One retractor, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	360 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-676 except that commercial colors are acceptable, or PPP-B-636, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Special requirements as specified in the Table II of PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements specified in Table I of PPP-B-636.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tracheal retractor is intended for use in surgical procedures for retraction of the trachea.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. National stock number.

6.3 National Stock Number (NSN) Coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Type
6515-00-361-8950	RETRACTOR, TRACHEAL, HUPP, 1 Sharp Prong	I
6515-00-361-8980	RETRACTOR, TRACHEAL, HUPP, 3 Sharp Prongs	II

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1750

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Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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